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Hello!
Basic Cellular System
Explore how a mobile network works
BTS
Mobile Station (MS)
PSTN (Other Networks)
HLR
VLR
AuC
EIR
Key Components
Mobile Station (MS)
User device (mobile phone)
Base Transceiver Station (BTS)
Provides radio communication
Base Station Controller (BSC)
Controls multiple BTSs
Mobile Switching Center (MSC)
Call switching and routing
PSTN / Other Networks
Connects external networks
Explore Components →
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Mobile Station (MS)
User device in a cellular system
Voice Calling
SMS Messaging
Mobile Data
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SIM Card
Antenna (Radio Interface)
Functions
Make and receive calls (voice communication)
Send and receive text messages (SMS)
Access internet and data services (mobile data)
Stores subscriber identity in SIM (IMSI, authentication keys)
Next: Base Station Subsystem →
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Base Station Subsystem
BTS and BSC
BTS (Cell 1)
BTS (Cell 2)
BTS (Cell 3)
BSC
BSC (Base Station Controller)
Base Transceiver Station (BTS)
Provides radio communication with MS
Handles transmission and reception
Covers a cell area
Base Station Controller (BSC)
Controls multiple BTSs
Allocates radio resources
Handles handover between cells
Interfaces with MSC
Next: Network Switching Subsystem →
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Network Switching Subsystem
Call switching, routing and subscriber management
HLR (Home Location Register)
Permanent subscriber data
VLR (Visitor Location Register)
Temporary subscriber data
MSC (Mobile Switching Center)
AuC (Authentication Center)
Authentication and security
EIR (Equipment Identity Register)
Validates mobile devices
Functions
MSC
Call switching, routing and mobility management
HLR
Stores permanent subscriber information (IMSI, location, services)
VLR
Stores temporary data for users in current area
AuC
Authenticates subscribers and generates security keys
EIR
Checks device identity (IMEI) and blocks invalid devices
Next: Communication Flow →
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End-to-End Communication
Call flow in a basic cellular system
1
Mobile Station (MS)
User makes a call
2
BTS
Receives radio signals from MS
3
BSC
Controls BTS and forwards the request
4
MSC
Routes the call and checks subscriber information (HLR/VLR)
5
PSTN / Other Networks
Connects to external network
6
Destination Phone
Call is established
Result
End-to-end communication is established between the two mobile stations.
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